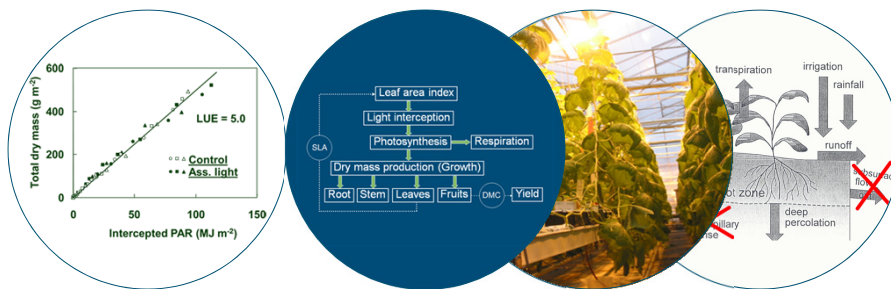


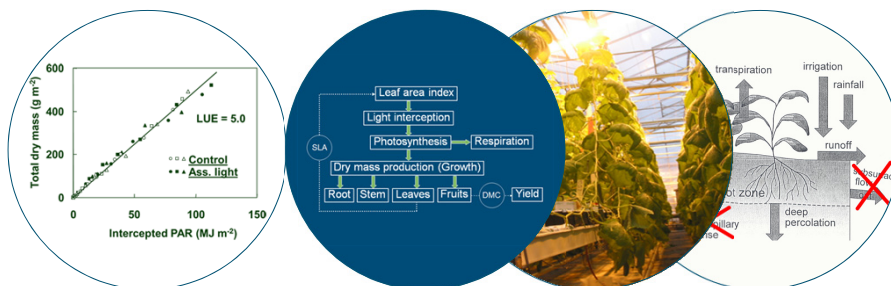
Lintul1: Potential crop growth

Dr. ir. Peter Leffelaar - Wageningen University

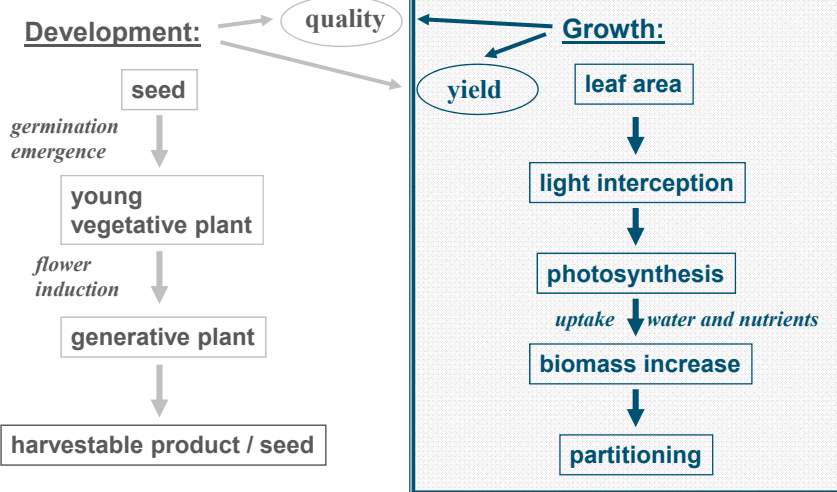


Lintul1: Potential crop growth

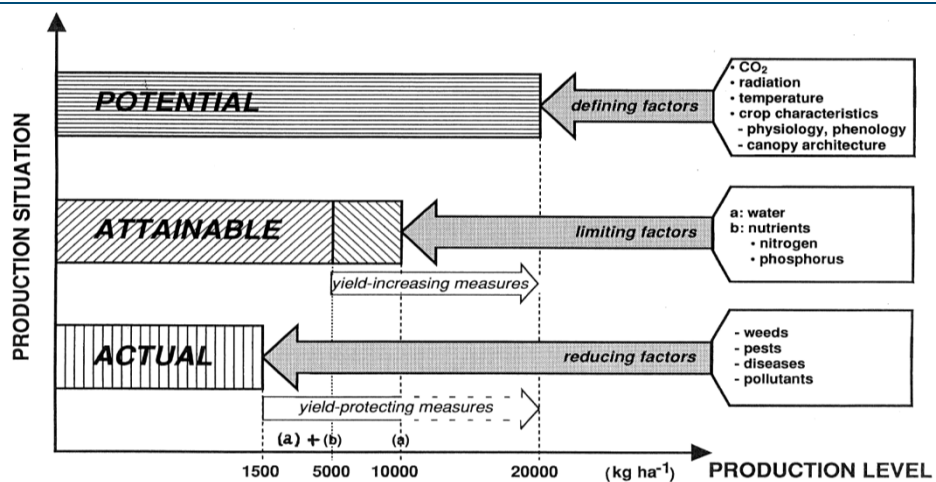
Introduction



Lintul1: Potential crop growth - scope of the course -

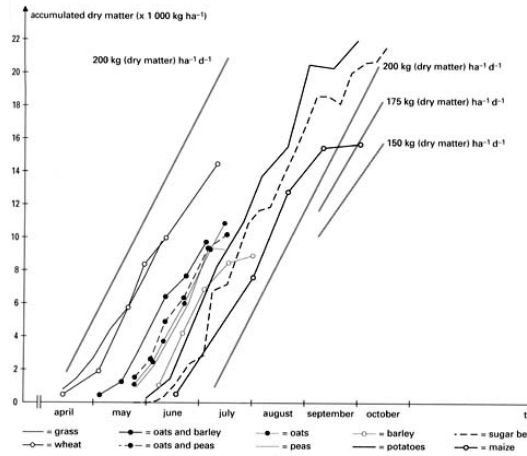


Potential, attainable and actual production levels



Schematic overview of production situations with corresponding dry matter production levels

Optimal growth rates - rule of thumb

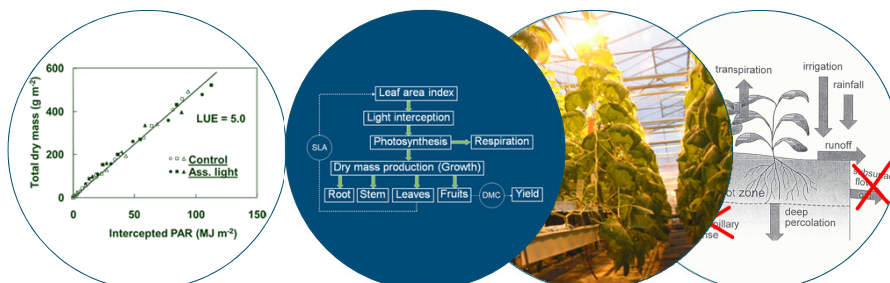


Growth rates of the main agricultural crops in the Netherlands under (near)-optimal growth conditions, compared to growth rates of 200, 175 and 150 kg (dry matter) ha⁻¹ d⁻¹
(Source: Sibma, 1968)

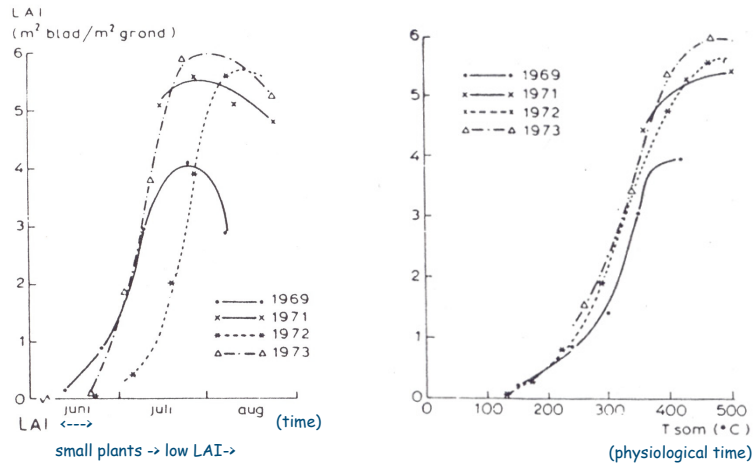
From: Sibma, 1968

Lintul1: Potential crop growth

Light and light interception



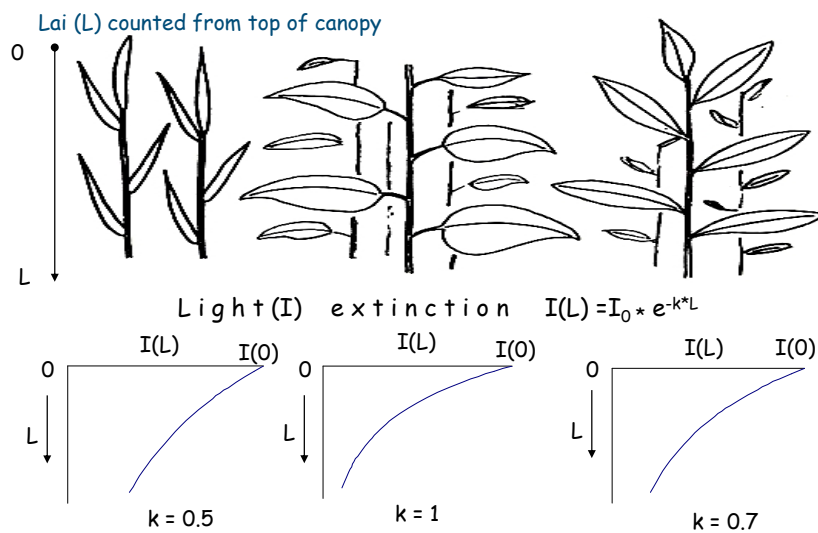
Leaf area of maize as a function of (physiological) time



small plants → low LAI →
not all sunlight used →
growth rate << 200 kg/(ha*d)

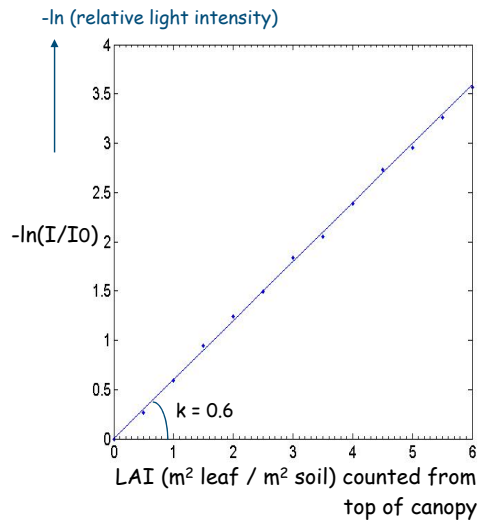
From: Sibma, 1987

Light extinction is a function of LAI



From: Coombs et al., 1985

Light extinction coefficient, k , follows from slope



$$I(L) = I_0 \cdot e^{-k \cdot L}$$

$$\Rightarrow -\ln\left(\frac{I}{I_0}\right) = k \cdot L$$

maize

From: Sibma, 1987

Spectral composition of global radiation

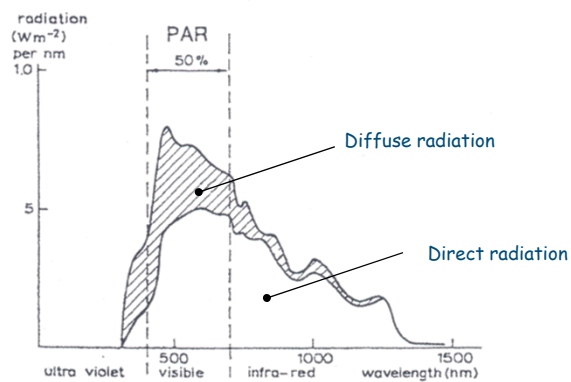
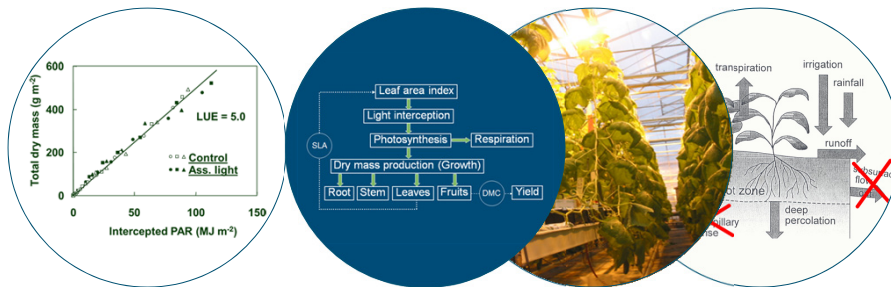


figure 2.2 Spectral composition of global radiation, discriminating between direct radiation (clear area) and diffuse (i.e. indirect) radiation scattered by clouds (hatched area).
 PAR = photosynthetically active radiation.
 (Source: Monteith and Unsworth, 1990, Principles of environmental physics. Edward Arnold, London, 291 p.)

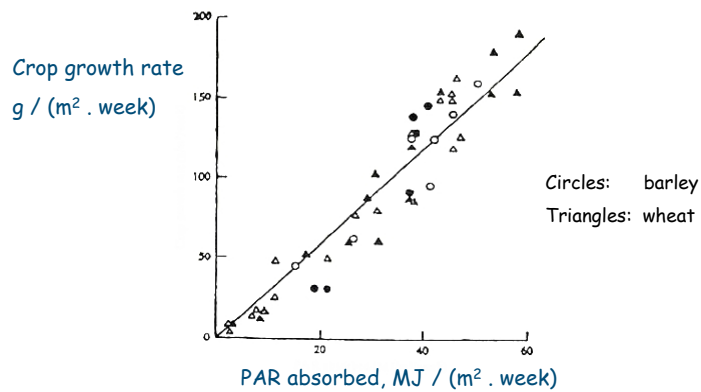
From: Monteith & Unsworth, 1990

Lintul1: Potential crop growth

Light (or Radiation) use efficiency



Light Use Efficiency, LUE, follows from slope



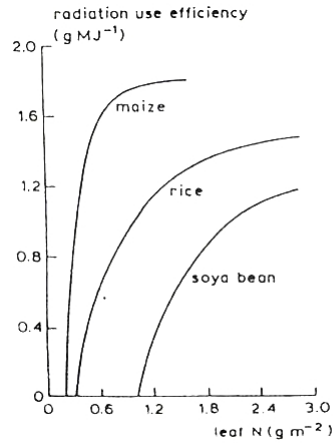
PARAM LUE = 3.0 (g Dry Matter per MJ PAR)

LUE Pinus = 1.9

LUE Quercus = 2.1

From: Gallagher & Biscoe, 1978

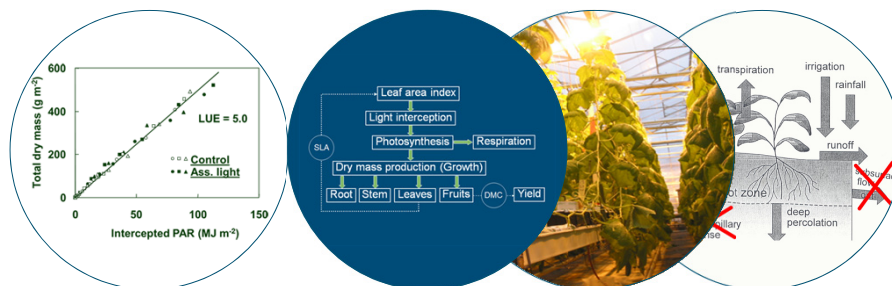
Radiation use efficiency as a function of leaf nitrogen content per unit area



From: Sinclair & Horie, 1989

Lintul1: Potential crop growth

Light interception, utilization and growth



Light interception, utilization and growth in LINTUL

TITLE LINTUL

INITIAL

INCON WTOTI = 5.

TIMER STTIME=1. ; FINTIM=100. ; DELT=1. ; PRDEL = 5.

TRANSLATION_GENERAL DRIVER = 'EUDRIV'

PRINT LAI, WTHA, PARINT

DYNAMIC

LAI = WTOTAL * LWR * SLA

PARINT = (0.5 * DTR) * (1. - EXP(-K * LAI))

$L = W F_{lv} S_{la}$ compare Eq.(13)
 $I_{int} = 0.5 DTR (1 - e^{-KL})$ Eq.(2)

GTOTAL = LUE * PARINT

WTOTAL = INTGRL(WTOTI, GTOTAL)

WTHA = WTOTAL / 100.

$dW/dt = RUE I_{int}$ Eq.(3)
 $W = \int dW/dt dt$ Eq.(4)

PARAM DTR = 20. ; LWR = 0.5 ; SLA = 0.022 ; LUE = 3.0 ; K = 0.6

END

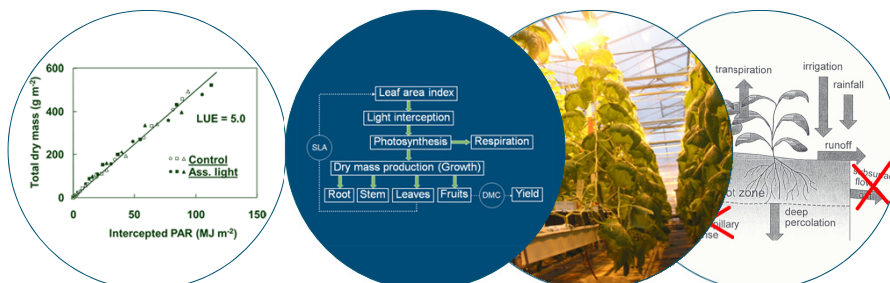
STOP

ENDJOB

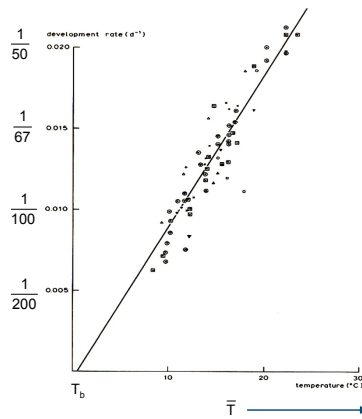


Lintul1: Potential crop growth

Crop development



Development rate of spring wheat as a function of temperature from emergence till flowering . . .



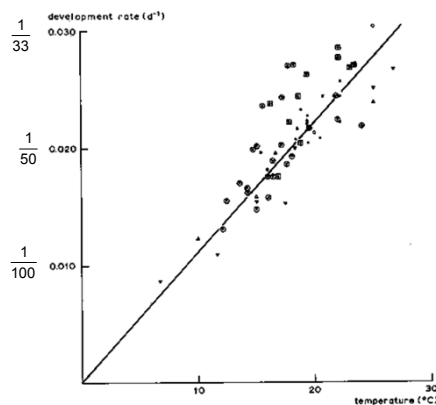
$$\frac{1}{\text{\# days}} = \bar{T} \cdot 0.0009 \Rightarrow$$

$$\text{"temperature sum"} = \bar{T} \cdot \text{\# days} \Rightarrow$$

$$\frac{1}{0.0009} = 1110 \text{ °Cd}$$

Data from over 20 papers assembled
by: Van Keulen & Seligman, 1987

. . . and from flowering to ripeness



$$\frac{1}{\text{\# days}} = \bar{T} \cdot 0.00103 \Rightarrow$$

$$\bar{T} \cdot \text{\# days} = \frac{1}{0.00103} = 970 \text{ °Cd}$$

Vegetative Phase: 1110 °Cd

Generative Phase: 970 °Cd

Whole growing season: 2080 °Cd

finish condition Lintul

Data from over 20 papers assembled
by: Van Keulen & Seligman, 1987

Use of weather data in LINTUL

TITLE LINTUL

* 1. Initial conditions and run control

FINISH **TSUM** > 2080.

* 2. Environmental data and temperature sum

WEATHER WTRDIR='C:\SYS\WEATHER'; CNTR='IT'; ISTN=3; ...
IYEAR=1985

DTR = RDD / 1.E+6

DAVTMP = 0.5 * (TMMN + TMMX)

DTEFF = MAX(0. , DAVTMP - TBASE)

EMERG = INSW(TIME-DOYEM , 0. , 1.)

TSUM = INTGRL (ZERO, RTSUM)

RTSUM = DTEFF * EMERG

$$T_{sum} = \int (T_a - T_b) dt$$

* 6. Functions and parameters for spring wheat

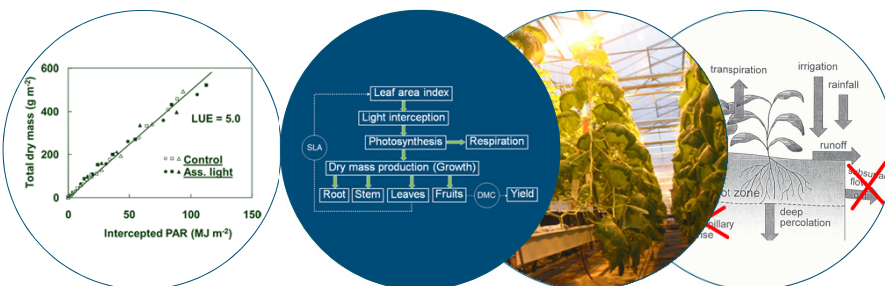
PARAM **TBASE** = 0.

PARAM DOYEM = 32.

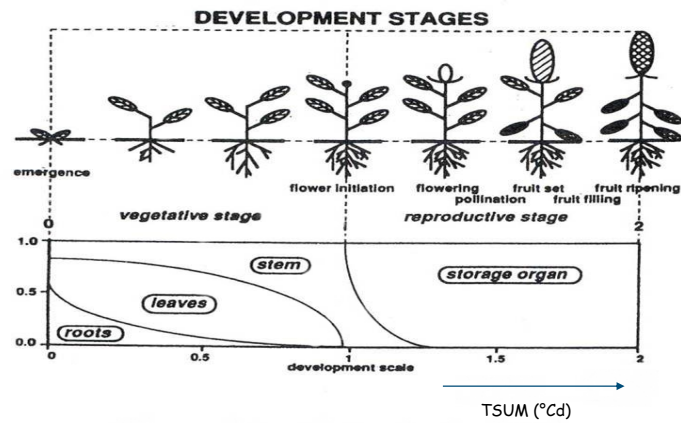


Lintul1: Potential crop growth

Development stages and allocation of dry matter

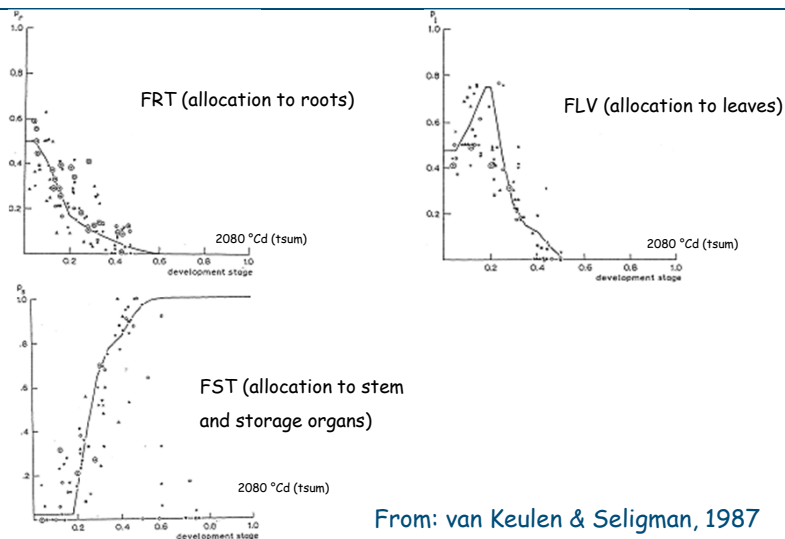


Crop development stages



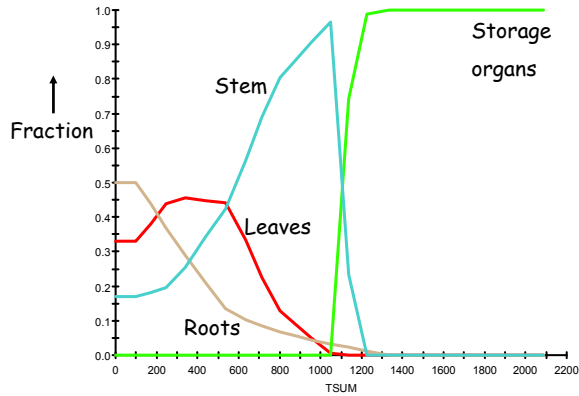
From: Lönstein, Lantinga, Rabbinge and van Keulen, 1995

Substrate allocation patterns in Spring Wheat: Measured



From: van Keulen & Seligman, 1987

Substrate allocation patterns in Spring Wheat: In model



Allocation pattern of assimilates for spring wheat
as used in Lintul (Light interception and utilization)



Growth rates and dry matter production of plant organs

TITLE LINTUL

FRT = AFGEN(FRTTB , Tsum)
FLV = AFGEN(FLVTB , Tsum)
FST = AFGEN(FSTTB , Tsum)
FSO = AFGEN(FSOTB , Tsum)

WLVG = INTGRL(WLVI , RWLVG)
WLVD = INTGRL(ZERO , DLV)
WST = INTGRL(ZERO , RWST)
WSO = INTGRL(ZERO , RWSO)
WSO_{THA} = WSO / 100.
WRT = INTGRL(ZERO , RWRT)
WLV = WLVG + WLVD

RWLVG = GTOTAL * FLV - DLV
RWST = GTOTAL * FST
RWSO = GTOTAL * FSO
RWRT = GTOTAL * FRT

$$dW_{lvg} / dt = RUE I_{int} F_{lv} - r_d W_{lvg} \quad \text{Eq.(8)}$$

$$dW_{st} / dt = RUE I_{int} F_{st} \quad \text{Eq.(7)}$$

$$dW_{so} / dt = RUE I_{int} F_{so} \quad \text{Eq.(7)}$$

$$dW_{rt} / dt = RUE I_{int} F_{rt} \quad \text{Eq.(7)}$$



Functions for spring wheat

TITLE LINTUL

FUNCTION FRTTB = 0.,0.50 , 110., 0.50 , ...
 275.,0.34 , 555.,0.12 , 780.,0.07 , ...
 1055.,0.03 , 1160.,0.02 , 1305.,0. , 2500.,0.

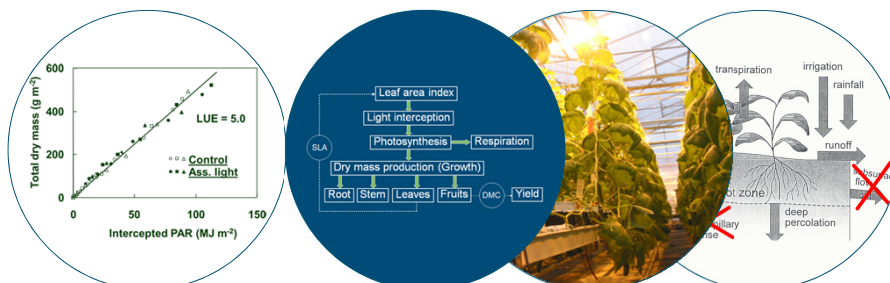
FUNCTION FLVTB = 0.,0.33 , 110.,0.33 , ...
 275.,0.46 , 555.,0.44 , 780.,0.14 , ...
 1055.,0. , 2500.,0.

FUNCTION FSTTB = 0.,0.17 , 110.,0.17 , ...
 275., 0.20 , 555., 0.44 , 780., 0.79 , ...
 1055., 0.97 , 1160., 0. , 2500., 0.

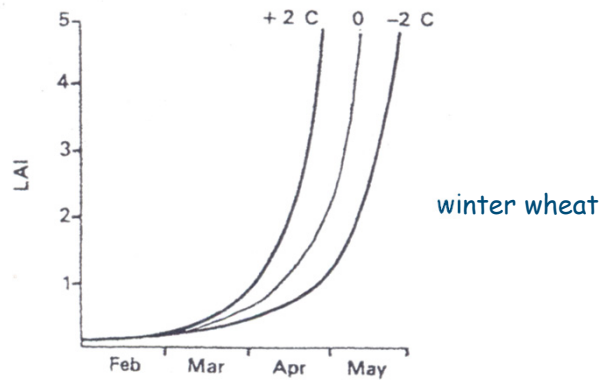
FUNCTION FSOTB = 0., 0. , ...
 1055.,0. , 1160.,0.98 , 1305.,1. , 2500.,1.

Lintul1: Potential crop growth

Early leaf growth



Early leaf growth: temperature dependent



From: Monteith, 1966

Parameters for spring wheat & Growth of Leaf Area (GLA)

```

PARAM LAII = 0.012 ; SLA = 0.022 ; RGRL = 0.009 ; ...
      TSUMAN = 1110. ; LAICR = 4. ; RDRSHM = 0.03
FUNCTION RDRT = -10,0.03 , 10,0.03 , 15,0.04 , 30,0.09 , 50,0.09
*-----*

```

```

      SUBROUTINE GLA(TIME,DOYEM,DTEFF,TSUM,LAII,RGRL,DELT,
$      SLA ,LAI,GLV,
      IMPLICIT REAL (A-Z)
      GLAI)

```

*---- Growth during **maturation** stage:

$$GLAI = SLA * GLV$$

$$dL/dt = S_{lo} F_{lv} RUE I_{int}$$

Eq.(7 & 13)

*---- Growth during **juvenile** stage:

IF ((TSUM.LT.330.) .AND. (LAI.LT.0.75))

$$GLAI = LAI * (EXP(RGRL * DTEFF * DELT) - 1.) / DELT$$

Eq.(12)

*---- Growth at day of seedling emergence:

IF ((TIME.GE.DOYEM) .AND. (LAI.EQ.0.))

$$GLAI = LAII / DELT$$

*---- Growth before seedling emergence:

IF (TIME.LT.DOYEM) GLAI = 0.

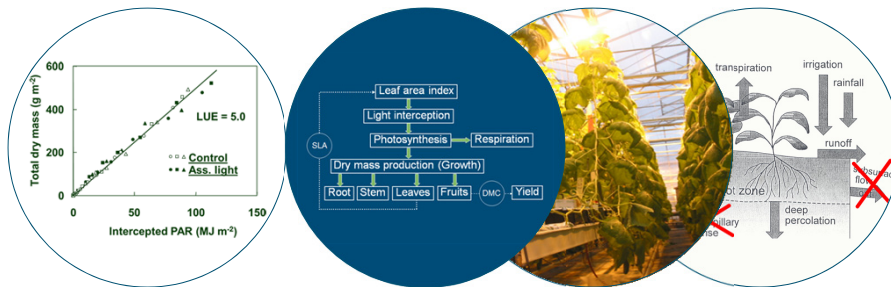
```

      RETURN
      END

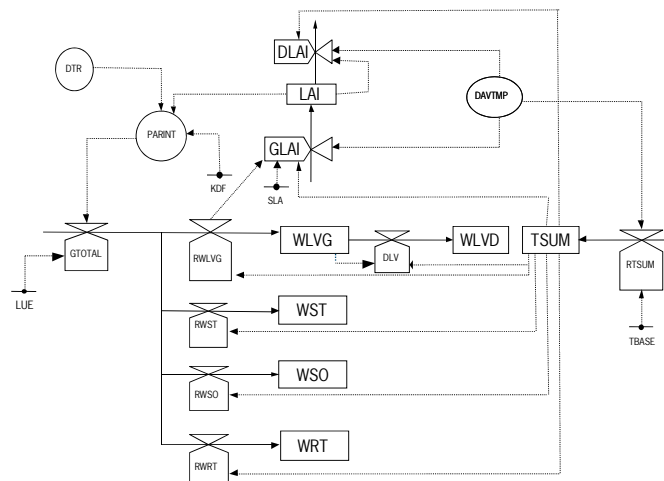
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Lintul1: Potential crop growth

Summarizing relational diagram and meaning symbols



Relational diagram of Lintul1

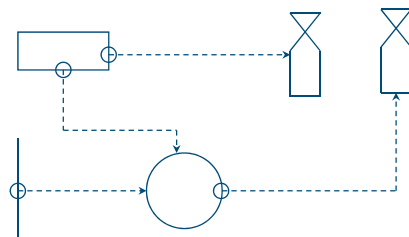


Forrester notation . . .

Symbol		Dimension
	Integral	Amount
	Differential equation (decision about rate)	Amount/time
	Flow of substance	Amount/time
	Discontinuous flow of substance	Amount/time
	Flow of information	No unit
	Input variable	Not known beforehand
	Auxiliary variable	Not known beforehand

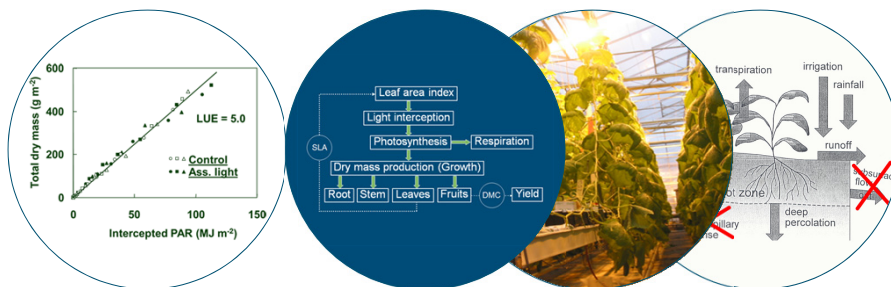
. . . Forrester notation (continued)

Information flows can only be used:
 Information flows should point towards a
 place of decision, never to a state:



Lintul1: Potential crop growth

Exercises



Practical exercises Lintul1

Exercises on the following subjects:

- Consequence of weather and different emergence dates on crop growth
- The sensitivity of yield for some important crop parameters or, differently stated, the consequence of uncertainties in parameter values
- Effects of parameter changes in light extinction (light capture), and light use efficiency on growth
- Potential yield as a function of emergence dates

There are data provided for a dry, an average and a wet year

Choose between Brazil and the Netherlands

Prepare a report for yourself on the PC to prepare for the exam

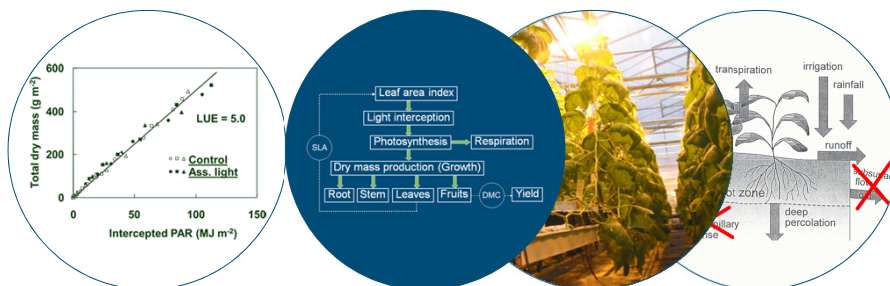


To be able to do the exercises, study:

- Theory: "Lintul-1: potential crop growth"
- This power point presentation

Lintul1: Potential crop growth

References



References used in the slides

- Coombs, J., D.O. Hall, S.P. Long and J.M.O. Scurlock, 1985. Techniques in bioproductivity and photosynthesis. Pergamon International Library of science, Technology, engineering and social studies, Oxford, pp.298.
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- Van Keulen, H. & N.G. Seligman, 1987. Simulation of water use, nitrogen nutrition and growth of a spring wheat crop. *Simulation Monographs*, Pudoc, Wageningen, pp. 310.